



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
Final Year B.Tech. Civil Engineering
W.E.F. 2026-27

Department of Civil Engineering

Vision

- To excel and lead in education, research, and innovation, contributing to the advancement of design, construction, and maintenance of infrastructure, to enhance the quality of life for humanity in a sustainable way.

Mission

- **M1:** To provide an outstanding learning experience through a rigorous curriculum of theory and practice that develops students' technical and professional skills to succeed in a wide range of careers.
- **M2:** To continually advance research through a culture of discovery, creativity, and innovation to benefit humankind.
- **M3:** To serve as highly capable resources to society, and the profession through professional organizations, consultancy, and continuing education.



Department of Civil Engineering

Program Educational Objectives (PEOs)

- Graduate will demonstrate peer-recognized technical competency in the analysis, design and construction of Civil Engineering Structures.
- Graduates will demonstrate leadership and initiative to advance professional and organizational goals with a commitment to ethical standards of profession, teamwork and respect for diverse cultural backgrounds.
- Graduates will be engaged in ongoing learning and professional development through pursuance of higher education and self-study.
- Graduates will be committed to create practice of engineering and other professions in a responsible manner contributing to the socio-economic development of the society.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative



	approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior, and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability, etc., with mutual understanding and respect, and of inclusive attitudes.
Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make



	effective presentations considering cultural, language, and learning difference
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO 11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)
Program Specific Outcomes (PSOs)	
1. Graduates will be able to survey, conduct geotechnical investigations, plan, analyze, design, estimate, and construct residences, public buildings, industrial buildings, townships, and infrastructural projects by adopting appropriate construction methods. 2. Graduates will be able to analyze and design the water resources systems, and municipal and industrial waste treatment plants with due consideration to a pollution-free environment. 3. Graduates will be able to use appropriate application software, develop skills necessary for professional practice as a Civil Engineer, and prepare themselves for competitive examinations for higher education & for public service commissions	



Department of Civil Engineering

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology



Department of Civil Engineering

Course Code Format

2	3	C	E	U/P	2	C	C	1	T/L
Year of Syllabus revision	Program Code	U-Under Graduate P-Post Graduate		Semester No./ Year1/2/3/...8	Course Type	Course Serial No1-9			T-Theory, L-Lab session A-Tutorial P-Programming / Design

Program Code

CE	Civil Engineering
----	-------------------

Course Type

BS	Basic Science
----	---------------

ES	Engineering Science
----	---------------------

HU	Humanities & Social Science
----	-----------------------------

MC	Mandatory Course
----	------------------

CC	Core Compulsory Course
----	------------------------

SN*	Self-Learning <i>N* indicates the serial number of electives offered in the respective category</i>
-----	--

EN*	Core Elective <i>N* indicates the serial number of electives offered in the respective category</i>
-----	--

ON*	Open Elective <i>N* indicates the serial number of electives offered in the respective category</i>
-----	--

SK	Skill Based Course
----	--------------------

SM	Seminar
----	---------

MP	Mini project
----	--------------

PR	Project
----	---------

IN	Internship
----	------------

Sample Course Code

23CEU7CC1T	Project Management
------------	--------------------



Department of Civil Engineering

B. Tech. Semester VII

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/POE	ISE	ICA	
23CEU7CC1T	Project Management	3	-	-	3	60		40	-	100
23CEU7CC2T	Advanced Design of Concrete Structures	3	-	-	3	60	-	40	-	100
23CEU7CC2A	Advanced Design of Concrete Structures	-	1	-	1	-	-	-	25	25
23CEU7EN*3T	Program Elective Course IV	2	-	-	2	60	-	40	-	100
23CEU7EN*4T	Program Elective Course V	2	-	-	2	60	-	40	-	100
23##U7MD5T	Multidisciplinary Minor V	3	-	-	3	60	-	40	-	100
23##U7MD5A	Multidisciplinary Minor V	-	1	-	1	-	-	-	25	25
Subtotal		13	2	-	15	300	-	200	50	550
Laboratory Courses										
23CEU7EN*3L	Program Elective Course IV	-	-	2	1	-	-	-	25	25
23CEU7EN*4L	Program Elective Course V	-	-	2	1	-	-	-	25	25
23CEU7CC6L	RCC Structural Design and Drawing	-	-	2	1	-	25	-	25	50
23CEU7PR7L	Project Phase I	-	-	8	4		50	-	100	150
Subtotal		-	-	14	7	-	75	-	175	250
Grand Total		13	2	14	22	300	75	200	225	800

Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of offering Programme

Department of Civil Engineering										
B. Tech. Semester VIII										
Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/POE	ISE	ICA	
23CMU8RN*1T	Research Methodology and Intellectual Property Rights/ MOOC	-	-	-	2	50	-	-	-	50
23CEU8EN*2T	Program Elective Course VI / MOOC#	-	-	-	2	50	-	-	-	50
	Subtotal	4	-	-	4	100	-	-	-	100
Laboratory Courses										
23CEU8IN3L	Internship	-	-	20	10	-	50	-	100	150
23CEU8PR4L	Project Phase II	-	-	4	2	-	50	-	100	150
	Subtotal	-	-	24	12	-	100	-	200	300
	Grand Total	4	-	24	16	200	-	200	400	

- N* indicates the serial number of electives offered in the respective category
- # List of Courses will be provided by the BoS.

• **List of Program Elective IV**

List of Program Elective Course -IV (Semester –VII)	
Course Code	Course title
23CEU7E13T	Pavement Materials and Design
23CEU7E23T	Ultra-High-Performance Concrete
23CEU7E33T	Repairs and Rehabilitation of Structures

- **List of Program Elective V**

List of Program Elective Course – V (Semester –VII)	
Course Code	Course title
23CEU7E14T	Earthquake-Resistant Design of RC Structures
23CEU7E24T	Urban Transportation Planning
23CEU7E34T	Environmental Audits & Industrial Compliance

- **List of Program Elective VI**

List of Program Elective Course -VI (Semester –VIII)	
Course Code	Course title
23CEU8E12T	Geosynthetics and soil structures
23CEU8E22T	Remote Sensing & GIS
23CEU8E32T	Planning and Design of Rural Roads

- **For Courses offered during Semester VIII**

Sr. No	Course Code	Option -I	Option II
1	23CMU8RN*1T	‘Research Methodology & IPR’ course offered by the department during Semester VIII in Self-Learning / Online mode, and successfully passing ESE for the same. 23CMU8R11T – Self Learning Mode	MOOC Certification* 23CMU8R21T – For MOOC (Option I) 23CMU8R31T – For MOOC (Option II)
2	23CEU8EN*2T	‘Program Elective VI’ course offered by the department during Semester VIII in Self-Learning / Online mode, and successfully passing ESE for the same. Program Elective VI options: 23CEU8E12T- Geosynthetics and soil structures 23CEU8E22T-Remote Sensing & GIS 23CEU8E32T-Planning and Design of Rural Roads	MOOC Certification* 23ETU8E42T – MOOC (Option I) 23ETU8E52T – MOOC (Option II) 23ETU8E62T – MOOC (Option III)

- If students have not completed the MOOC in Semester V or VI, then students have to complete this MOOC in the Semester VII or VIII. In Semester VIII, the marks earned by the

students in the final assessment of this MOOC will be appropriately scaled and transferred for 23CMU8RN*1T and/or 23CEU8EN*2T, along with the required credits. The list of approved MOOCs will be finalized and released by the Board of Studies before the commencement of the semester, as per the available courses on the MOOC platform.

- If students do not pass the MOOC examination of 23CMU8RN*1T or 23CEU8EN*2T before commencement of the End Semester Examination (ESE) of Semester VIII, then they must appear for the institute-conducted ESE for courses (mentioned in option I) by the department during Semester VIII.
- **Multidisciplinary Minor (MDM) Courses**

Sr. No.	MDM Program	MDM I (Sem III)	MDM II (Sem IV)	MDM III (Sem V)	MDM IV (Sem VI)	MDM V (Sem VII)
1	Computer Science and Engineering	Operating System	UI Technologies	Software Engineering	Big Data Technologies	Software Testing and Quality Assurance
2	Information Technology	Principles of Operating Systems	Web UI and UX Technology	Software Engineering Principles	Data Preprocessing and Visualization	Cyber Security
3	Mechanical and Automation Engineering	Manufacturing Processes and Mechanisms	Machine Drawing and 3D Modelling	Automotive Engineering and Robotics	Additive Manufacturing	Thermal Systems
4	Electronics and Telecommunication Engineering	Fundamentals of Electronic Circuits	Electronics Design and Prototyping	Introduction to Embedded Systems	Fundamentals of Communication Techniques	Enclosure and Communication Design for IoT
5	Electronics and Computer Engineering	Analog Electronics	Digital Electronics	Microcontrollers and Peripherals	Advanced Controllers and Interfacing	Electronic System Design
6	Artificial Intelligence and Machine Learning	Fundamentals of Machine Learning	Data Science and Visualization	Reinforcement Learning	Applied Machine Learning	Generative AI



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7CC1T: Project Management

Teaching Scheme

Lectures 3 Hours/week

Credits 3

Examination Scheme

ESE 60 Marks

ISE 40 Marks

Introduction:

This course provides an application of the Project Management Tools and techniques required for decision-making related to the project for successful completion. The course introduces Construction Safety, Construction equipment, and advanced technology required on Civil Engineering Sites. Additionally, the course covers Project Management software and its application for Civil Engineering Projects.

Course Prerequisite:

The student has completed a course, Principles of Management and Quantitative Techniques, and has acquired knowledge of various management principles, management functions, and Quantitative techniques, including decision-making methods. The student is also familiar with inventory management and quality control requirements for Civil Engineering Projects.

Course Objectives:

1. To make students learn about Project management tools, such as the Bar charts and Network diagrams, required for the successful completion of projects.
2. To make students learn Network Compression and Updating to make various decisions related to the projects.
3. To instruct students about various construction safety measures and methods in the construction industry.
4. To educate students about Construction Equipment and Advanced formworks.
5. To educate students about project management software for the successful completion of Civil Engineering Projects

Course Outcomes:

After completing the course, students will be able to

1. Plan the project and prepare Bar charts and Network diagrams for the declaration of project duration and critical path.
2. Carry out Network Compression to optimize the project duration and cost, and also update the project.
3. Establish appropriate safety measures for various Civil Engineering Projects
4. Select appropriate Construction Equipment and Advanced formworks for Civil Engineering projects.
5. Use appropriate project management application software for planning, tracking, and reporting the progress of civil engineering projects



Unit – I	Project Management, Network Diagram and Critical Path Method (CPM)	8 Hours
▪ Introduction, Steps in Project Management, Work Breakdown Structure (WBS). Gantt (Bar) Chart, Mile Stone chart. ▪ Development of network: Representation by Activity on Arrow (AOA) and Activity on Node (AON), Fulkerson’s Rule. ▪ Critical Path Method (CPM): Introduction, Time estimates, floats, critical path.		
Unit – II	Performance Evaluation and Review Techniques (PERT) and Precedence Network	6 Hours
▪ Concept of probability, Normal and Beta Distribution, Central limit theorem. Time estimates and calculations of project duration, critical path, slack, and probability of project completion. ▪ Precedence Network and example		
Unit – III	Network Compression and Updating	6 Hours
▪ Cost associated with Activity in Project: Least Cost and Optimum Duration. ▪ Resource allocation: Smoothing and levelling. ▪ Updating: Need, steps, project duration, and calculation for updated network.		
Unit – IV	Construction Safety	8 Hours
▪ Safety against accidents on various construction sites such as buildings, roads, tunnels, bridges, fabrication and erection works, etc. ▪ Safety at various stages of construction. ▪ Safety measures in construction.		
Unit – V	Construction Equipment and Advanced formworks	5 Hours
▪ Earthmoving Equipment - Tractors, Bulldozers, Scrappers, Power shovels, Hoes, Dragline. ▪ Hauling Equipment: Tipper (Dumper Truck) ▪ Compactors- types and performance, operating efficiencies. ▪ Lifting Cranes: Mobile and Floating Cranes. ▪ Advanced formworks- Aluform and Mivan Formwork		
Unit – VI	Project Management Software (PMS) and Building Information Modeling (BIM)	6 Hours
▪ Introduction to applications of PMS such as MS Excel, MS Project, Primavera, and PRINCE and Open-Source software. ▪ Introduction to BIM (Building Information Modeling)		
Text Books		
1. A Management Guide to PERT/CPM: Weist J. D. ,Levy, Prentice Hall of India, New Delhi, 2nd Ed. 1982 2. PERT and CPM Principles and Applications: Srinath L. S., East West Publication, New		



Delhi, 3rd Ed. 1995.

3. PERT and CPM- B. C. Punmia, K. K. Khandelwal, Laxmi Publications, New Delhi, 4th Ed. 2012.
4. Computerized Project Management Technique for Manufacturing and construction: Samaras T.T., Kim Yensueng, Prentice Hall of India, New Delhi, 1979.
5. Principles of Construction Management: Roy Pilcher , Tata McGraw Hill Publications.
6. Construction, Planning, Equipment and methods - R. L . Peurifoy McGraw hill book co New Delhi
7. Construction Equipment Guide, David A. Day, Neal B. H. Benjamin, John Wiley & Sons.
8. Construction Equipment – Mahesh Varma ,Metropolitan book co ,New York
9. Heavy Construction – Planning, Equipment and methods – Jagman Singh, Oxford and IBH publishers, New Delhi.
10. Construction of Diaphragm Walls, I Hajnal, I Marton, F. Regele Wiley Interscience Publication, John Wiley & Sons.
11. Structural & cut off Diaphragm walls, R.G.H. Boyes, Applied Science Publishers Ltd., London

Reference Books

1. CPM in Construction Practice, Antill J. M., John Wiley and Sons.
2. Construction Project Management – Planning, Scheduling and Control- Chitkara K.K., Tata McGraw Hill Publications New Delhi, 4th Ed. 2002.
3. Construction Planning and Management through System Techniques: Verma M., Metropolitan Publication, 3rd Ed. 1985.
4. Construction Project Management- Bennett J. M. Clough R. H., Butterworth's Wiley John, New Delhi, 1972.
5. Construction Scheduling with Primavera Enterprise- Marchman D.A., Thomson/Brooks- Cole.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7CC2T: Advanced Design of Concrete Structures

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Tutorial	1 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks
Introduction:			
Advanced Design of Concrete Structures: A Comprehensive Course Covering the Principles and Practices of Designing Various Concrete Elements. The curriculum includes the design of philosophies of LSM, analysis and design of singly and doubly reinforced T & L sections. This course covers principles of Prestressing, various types of prestressed sections, classifications, losses in prestress and design of prestressed members.			
Course Prerequisite:			
The student has completed courses such as Introduction to Solid Mechanics, Structural Analysis, and Design of Concrete Structures, demonstrating proficient knowledge in load calculation, shear force, and bending moments. They are also well-versed in the classifications of beam sections and their supports.			
Course Objectives:			
1. Impart knowledge of the design of flanged sections. 2. Enable selection and design of continuous beams for suitable boundary conditions. 3. Develop competence in the design of beams subjected to combined bending, shear & torsion. 4. Impart knowledge of the analysis of prestressed concrete system & losses in prestress. 5. Develop competence in the design of prestressed concrete members.			
Course Outcomes:			
After completing the course, students will be able to, 1. Design singly and doubly reinforced flanged sections (T & L beams). 2. Design continuous reinforced concrete beams for given loading and support conditions. 3. Design reinforced concrete beams subjected to combined bending, shear and torsion. 4. Analyze various prestressing systems & compute the losses in prestress in members 5. Design prestressed concrete beams of rectangular and symmetrical I section.			
Unit – I	Design of Flanged Sections	10 Hours	



Design of Singly and doubly Reinforced T & L Beams for flexure.		
Unit – II	Design of Continuous Beams	10 Hours
Design of Continuous Beams by Limit State Method		
Unit – III	Limit State of Collapse (Flexure, Shear & Torsion)	8 Hours
Behaviour of R.C. rectangular sections subjected to torsion, Design of sections subjected to combined bending and torsion, combined shear and torsion, Design of beams for torsion.		
Unit – IV	Prestressed concrete	8 Hours
Introduction to Prestress, methods, Stress concepts, Strength concept, Load Bearing concept, Thrust line, Various cable profiles.		
Unit – V	Losses in Prestress	6 Hours
Losses in prestress – Shrinkage, Creep, Elastic Shortening, Relaxation of steel, Curvature effect in Pre-tensioned & Post-tensioned prestress concrete members.		
Unit – VI	Design of Prestressed Concrete Beam	8 Hours
Design of Rectangular and Symmetrical I-Section.		
Note: - Use of IS: 456-2000, latest version is permitted in ISE and ESE examinations.		
INTERNAL CONTINUOUS ASSESSMENT (ICA)		
Internal Continuous Assessment (ICA) shall consist of a minimum of six assignments based on the entire curriculum, which will be assessed by faculty.		
Text Books		
1. Design of Steel Structures, by N. Subramanian, Oxford, 2008 2. Limit State Design of Steel Structures, by S. K. Duggal 3. Design of steel structure by Limit State Method as per IS: 800- 2007 by Bhavikatti S. K 4. International Publishing House, New Delhi 5. Limit state design in Structural Steel by Dr M. R. Shiyekar 6. Design of Steel Structures by K. S. Sai Ram 7. Design of Steel Structures by L. S. Jayagopal and D. Tensing		
Reference Books		
1. Limit state design of Steel Structure by V. L. Shah & Gore, Structures Publication, Pune 2. Limit State Design of Steel Structures by D. Ramchandra & Virendra Gehlot, Scientific Publishers		



3. Structural Design and Drawing Reinforced Concrete and Steel by N. Krishnaraju,
4. Universities Press (India) Pvt. Ltd. Hyderabad.
5. Bureau of Indian Standards, IS800-2007, IS875-latest. Steel Tables SP: 6(1) and SP: 6(6)

e-Resources

1. Web links and Video Lectures (e-Resources):
<https://nptel.ac.in/courses/105105162>
2. <https://youtu.be/6EklcOsA5VI?si=n0leGJxA46WiNNtz>
3. <https://youtu.be/g9NOsOhUgn8?si=czuxGmf3thNpgqjz>
4. <https://youtu.be/mgNeGbtcvZ0?si=AOFE7loUx-qiPcTW>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7E13T: Program Elective Course IV-Pavement Materials and Design

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course provides a thorough introduction to the fundamental materials utilised in pavement construction, encompassing soils, aggregates, bituminous binders, and concrete. Additionally, it covers the classification of bituminous binders and their application in recommending appropriate binders for road construction.

Course Prerequisite:

The student should have completed the Basics of Civil Engineering course, acquiring fundamental knowledge of aggregates, bituminous binders, and their applications in pavement construction, as well as an understanding of concrete and bituminous mix design, weather effects on pavements, and relevant testing methods.

Course Objectives:

1. To introduce students to soil as a highway material, its classification, engineering properties, and suitability for pavement applications.
2. To provide a comprehensive understanding of aggregates, including their origin, properties, gradation, and role in pavement performance.
3. To impart fundamental and advanced knowledge of bitumen and modified bituminous binders, their properties, tests, and field applications.
4. To familiarise students with bituminous mixtures, including production methods, volumetric concepts, and mix design procedures.
5. To develop competence in the design of flexible and rigid pavements using Indian and international pavement design methodologies.

Course Outcomes:

After completing the course, students will be able to

1. Classify soils and evaluate their engineering properties, such as strength, stiffness, and bearing capacity, for use as pavement subgrade materials.
2. Analyse aggregate characteristics including gradation, shape, strength, and mineralogy to assess their suitability for pavement construction.



3. Evaluate the physical, rheological, and ageing properties of bitumen, including modified binders, emulsions, and cutbacks, using standard laboratory tests. 4. Design bituminous mixes using Marshall and Superpave methods and assess their performance through volumetric and mechanical parameters. 5. Design flexible and rigid pavements using guidelines of the Indian Roads Congress and the AASHTO method, including evaluation of overlays and pavement performance.		
Unit – I	Soil	5 Hours
Introduction to soil as a highway material; Classification of soils; Consistency Limits; Soil compaction and role of moisture; Mechanical properties of soil (Shear strength, Unconfined compressive strength, Resilient modulus, California bearing ratio, Modulus of subgrade reaction etc.); Introduction to expansive soils, relevant tests, and soil stabilisation techniques.		
Unit – II	Aggregates	5 Hours
Aggregate origin, types, production, and quarrying operation; Classification of aggregates; Aggregate gradation and gradation parameters; Theories of aggregate blending; Minerology of aggregates and its importance; Aggregate shape and texture: quantification and importance; Aggregate strength properties, and relevant tests.		
Unit – III	Bitumen, Modified bitumen, Bitumen emulsion and Cutback bitumen	6 Hours
Bitumen as a binding agent; Production of bitumen; Physical and rheological properties of bitumen; Introduction to viscoelasticity; Chemistry of bitumen; Ageing of bitumen; Grading of bitumen, and relevant tests: Penetration grade, Viscosity grade, Performance grade; Bitumen modification: Need, Types and Importance; Introduction of bitumen emulsion: Theory of emulsification, Uses, Grading of emulsions, and Relevant tests; Introduction to cutback bitumen: Types, Uses, and relevant tests.		
Unit – IV	Bitumen Mixtures	6 Hours
Production of bituminous mixtures: Laboratory and Plant; Role of bituminous mixture and desirable properties; Volumetrics of bituminous mixture; Mix design of bituminous mixture: Marshall and Superpave methods; Mechanical tests and characterisation of bituminous mixtures; Introduction to performance-based mix design concepts; Mix design of cold bituminous mixtures; Mix design of hot recycled mixtures		
Unit – V	Pavement Design	10 Hours
Design of Flexible Pavements, Factors Affecting Design, IRC guidelines for Flexible Pavement Design, Deflection Studies in Flexible Pavements, Overlay Design as per IRC guidelines, Design of Rigid Pavements, Factors Affecting Design, Design of Rigid Pavements as per IRC guidelines, AASHTO method of Flexible and Rigid Pavement Design.		
INTERNAL CONTINUOUS ASSESSMENT (ICA) Internal Continuous Assessment (ICA) shall consist of a minimum of six assignments based on the entire curriculum, which will be assessed by faculty.		



Text Books

1. Principles of Transportation Engineering, Partha Chakroborty and Animesh Das, *Principles of Transportation Engineering*, 1st Edition, PHI Learning Private Limited, New Delhi, India.
2. Highway Engineering, S. K. Khanna, C. E. G. Justo and A. Veeraragavan, *Highway Engineering*, 10th Revised Edition, Nem Chand & Bros., Roorkee, India.
3. Pavement Design and Materials, A. T. Papagiannakis and E. A. Masad, *Pavement Design and Materials*, 2nd Edition, John Wiley & Sons, Hoboken, USA.
4. Pavement Engineering, Sandipan Goswami, *Pavement Engineering: Design, Construction and Maintenance*, 1st Edition, PHI Learning Private Limited, New Delhi, India.
5. Bituminous Road Construction in India, Prithvi S. Kandhal, A. Veeraragavan and Rajan Choudhary, *Bituminous Road Construction in India*, **2nd Edition**, McGraw Hill Education (India) Pvt. Ltd., New Delhi, India.

Reference Books

1. Pavement Engineering: Principles and Practice Rajib B. Mallick and Tahar El-Korchi, *Pavement Engineering: Principles and Practice*, 3rd Edition, CRC Press (Taylor & Francis Group), Boca Raton, USA.
2. Highway Engineering: Pavements, Materials and Control of Quality Athanassios Nikolaides, *Highway Engineering: Pavements, Materials and Control of Quality*, 1st Edition, CRC Press, London, UK.
3. Hot Mix Asphalt Materials, Mixture Design, and Construction E. R. Brown, P. S. Kandhal, F. L. Roberts, Y. R. Kim, D. Y. Lee and T. W. Kennedy, *Hot Mix Asphalt Materials, Mixture Design, and Construction*, 2nd Edition, National Asphalt Pavement Association (NAPA), Lanham, USA.
4. Transportation Engineering Volume I C. Venkatramaiah, *Transportation Engineering – Volume I*, 1st Edition, Universities Press, Hyderabad, India.
5. The Shell Bitumen Handbook, John Read and Duncan Whiteoak, *The Shell Bitumen Handbook*, 6th Edition, ICE Publishing, London, UK.

e-Resources

1. NPTEL, *Pavement Materials* and *Pavement Construction Technology* courses – Video lectures and downloadable notes by IIT faculty.
2. SWAYAM, Courses on *Highway Engineering* and *Transportation Engineering* aligned with IRC practices.
3. Coursera. Courses on *Pavement Engineering* and *Transportation Infrastructure* by international universities.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7E23T: Program Elective Course IV-Ultra High-Performance Concrete

Teaching Scheme

Lectures	2 Hours/week
Practical	2 Hours/week
Credits	3

Examination Scheme

ESE	60 Marks
ISE	40 Marks
ICA	25 Marks

Introduction:

This course provides an in-depth understanding of Ultra-High-Performance Concrete (UHPC), focusing on its production principles, material properties, durability, and mix design. It emphasises practical applications and real-world case studies, enabling learners to design and evaluate UHPC for advanced and sustainable infrastructure systems.

Course Prerequisite:

The student should have completed the Basics of Building Materials, Concrete Technology, Strength of Materials, and basic knowledge of durability and sustainability in construction materials

Course Objectives:

1. Introduce the fundamentals, evolution, and necessity of Ultra-High-Performance Concrete.
2. Explain the scientific principles governing the production of UHPC.
3. Understand fresh and hardened properties of UHPC and their characterization.
4. Evaluate durability performance and long-term behavior of UHPC.
5. Develop skills in UHPC mix design and optimization.
6. Familiarize students with real-world applications and case studies of UHPC in infrastructure projects.

Course Outcomes:

After completing the course, students will be able to

1. Explain the fundamental concepts, evolution, and significance of UHPC.
2. Apply production principles and material science concepts to UHPC development.
3. Evaluate fresh and hardened properties of UHPC using standard test methods.
4. Analyze durability performance and long-term behavior of UHPC.
5. Design and optimize UHPC mixes for specific structural applications.



Unit – I	Introduction	5 Hours
Background, Definition of Ultra High-Performance Concrete, Economic Factors, Raw Materials and Performance Characteristics, Manufacturing Considerations, Global UHPC codes, Standards, Specifications and Guidelines		
Unit – II	Principles for the Production of UHPC	5 Hours
Basic Material Concepts, Microstructure properties, Grading Optimisation, raw materials, Cement, Mineral Admixtures, Chemical Admixtures, Fine Aggregates, Water, Chloride Content, Fibres, Effect of Fibre content, Steel Fibres, Polymer Fibres, UHPC without Fibres and with Fibres. Mixing techniques and curing regimes (thermal curing, steam curing)		
Unit – III	Properties of Fresh and Hardened Concrete	6 Hours
Workability, Flow Table Test, Slump Flow Tests, Segregation resistance and fibre dispersion, Compressive Strength, Deformation, Fatigue, Durability, High Early strength, Modulus of Elasticity and Poisson's Ratio, Coefficient of Thermal Expansion, Tensile Strength, Bond Strength, Creep and Shrinkage. Stress–strain relationship		
Unit – IV	Durability Properties	6 Hours
Absorption, Permeability, Acid Resistance, Abrasion, Alkali-Silica Reaction (ASR), Carbonation, Cover, Spacing of Reinforcement, Freeze–thaw resistance, Fire resistance and thermal stability, Service life prediction and life-cycle assessment		
Unit – V	Mix Design	10 Hours
General aspects of Mix design, Differences between conventional concrete and UHPC mix design, Empirical and analytical mix design approaches, Optimization of binder content and particle size distribution, Fiber dosage optimization, Trial mixes and performance-based design, Quality control and field production challenges		
Unit – VI	Applications of UHPC and Case Studies	10 Hours
Applications of UHPC in bridge decks, girders, and joints, Pavement Construction, Precast structural elements, Repair and retrofitting applications, Architectural and façade elements, Case studies: UHPC bridges (global and Indian context), Pavement overlays and urban infrastructure, Industrial and defence structures.		



INTERNAL CONTINUOUS ASSESSMENT (ICA) Internal Continuous Assessment (ICA) shall consist of a minimum of six assignments based on the entire curriculum, which will be assessed by faculty.	
Text Books	
1.	Ultra-High-Performance Concrete UHPC – E. Fehling, M. Schmidt, J. Walraven
2.	Concrete Technology – M.S. Shetty
Reference Books	
1.	High Performance Concrete – P.K. Mehta and P.J.M. Monteiro
2.	Design and Construction of Concrete Structures – fib Model Code
3.	Advanced Concrete Technology – Zongjin Li
4.	AFGC Interim Recommendations for UHPC
5.	ASTM Standards on UHPC Testing and Materials
e-Resources	
1.	NPTEL Video Lectures on Advanced Concrete and High-Performance Materials
2.	RILEM Technical Committee reports on UHPC
3.	fib Bulletins on UHPC design and applications





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B. Tech. (Civil Engineering), Semester-VII

23CEU7E33T: Program Elective Course-IV - Repairs and Rehabilitation of Structures

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course provides a comprehensive understanding of maintenance and repair strategies for concrete structures with emphasis on serviceability and durability considerations. It introduces commonly used repair materials and retrofitting techniques for structural rehabilitation. The course also covers the fundamentals of structural health monitoring and outlines appropriate demolition techniques for deteriorated and dilapidated structures.

Course Prerequisite:

Students should complete a course in the Basics of Civil Engineering and Concrete Technology.

Course Objectives:

1. To introduce students to the principles, necessity, and significance of maintenance and repair strategies for concrete structures.
2. To familiarize students with serviceability and durability considerations affecting the performance and lifespan of concrete structures.
3. To provide comprehensive knowledge of materials, methods, and techniques used in the repair of structural elements.
4. To develop the ability to identify, evaluate, and select appropriate repair and retrofitting techniques for existing structures.
5. To create awareness of structural health monitoring systems and suitable demolition techniques for deteriorated and dilapidated structures.

Course Outcomes:

After completing the course, students will be able to

1. Explain the principles and importance of maintenance and repair strategies for concrete structures.
2. Analyze serviceability and durability aspects affecting the performance of concrete structures.
3. Identify and evaluate materials and techniques used for repairing structural elements.
4. Select and apply appropriate repair and retrofitting techniques for existing structures.
5. Evaluate and apply structural health monitoring techniques and appropriate demolition methods for deteriorated structures.



Unit – I	Assessment and Maintenance of structures:	5 Hours
Repair and rehabilitation, facts of maintenance, importance of maintenance various aspects of inspection, assessment procedure for evaluating damaged structure. Repair Strategies: Causes of distress in concrete structures, construction and design failures, Condition assessment and distress-diagnostic techniques.		
Unit – II	Serviceability and Durability of Concrete.	5 Hours
Quality assurance for concrete construction, concrete properties – strength, permeability, thermal properties and cracking. Effects due to climate, temperature, chemicals, corrosion, and durability test on concrete.		
Unit – III	Materials and Techniques for Repair	7 Hours
Special concretes and mortar, concrete chemicals, special elements for accelerated strength gain, expansive cement, polymer concrete, sulphur infiltrated concrete, ferro cement, fibre reinforced concrete, bacterial concrete, rust eliminators and polymers coating for rebars during repair, foamed concrete, mortar and dry pack, vacuum concrete, gunite and shotcrete, epoxy injection, mortar repair for cracks, shoring and underpinning.		
Unit – IV	Repair, Retrofitting, and Strengthening of Concrete and Heritage Structures	7 Hours
Repairs to overcome low member strength, deflection, chemical disruption, corrosion, fire, leakage-waterproofing, marine exposure, repairs in heritage structures. Strengthening of Structures: Strengthening Methods, retrofitting, jacketing.		
Unit – V	Health Monitoring and Demolition Techniques:	6 Hours
Long term health monitoring techniques, engineered demolition techniques for dilapidated structures, modern equipment, use of sensors for building instrumentation.		
ICA :- Internal Continuous Assessment (ICA) shall consist of a minimum of five assignments based on the entire curriculum, which will be assessed by faculty.		
Text Books		
1."Concrete Technology, Theory and Practice" by M.S. Shetty, S, Chand Publications, New Delhi 2. Rehabilitation of Concrete Structures by Dr.Vidivelli, Standard Publishers Distributors. 3. "Maintenance, Repair & Rehabilitation & Minor works of Buildings", by P. C. Varghese, PHI Learning Private Limited Delhi 4. "Repair and Rehabilitation of Concrete Structures", by Poonam Modi and Chirag Patel , PHI Learning Private Limited, Delhi. 5. "Maintenance, Repair, Rehabilitation, and Retrofitting of Structures", by Dr. G.Nandini Devi, WILEY Publications 6. "Repairs and Rehabilitation of Structures", by Dr. R.N Krishna , and Prof. A.R.Sanathakumar, Rajsons Publications Pvt. Ltd.		
Reference Books		



1. Concrete Technology by A.R. Santhakumar, Oxford University press
2. Defects and Deterioration in Buildings, E F & N Spon, London
3. Non-Destructive Evaluation of Concrete Structures by Bungey - Surrey University
4. Maintenance and Repair of Civil Structures, B.L. Gupta and Amit Gupta, Standard Publications.
5. Concrete Repair and Maintenance Illustrated, RS Means Company Inc W. H. Ranso, (1981).
6. Building Failures: Diagnosis and Avoidance, EF & N Spon, London,
7. Concrete- Microstructure, Properties and Materials, by Mehta, P.K and Montevic. P.J., ICI, 997
8. Civil Engineering Materials, Jackson, N., ELBS, 1983.

e-Resources

1. Web links and Video Lectures (e-Resources):
<https://nptel.ac.in/courses/105106202>
2. <https://www.nptelprep.in/courses/105106202/materials>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7E14T: Program Elective Course-V -Earthquake Resistant Design of RC Structures

Teaching Scheme		Examination Scheme	
Lectures:	2 Hours/week	ESE	60 Marks
Practical:	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Introduction:			
This course is designed to reflect the current developments and advances being made in the field of Earthquake-Resistant Design of Structures. This course facilitates the application of earthquake-resistant principles, features, and criteria to structures through the provisions of the latest IS 1893 & 13920.			
Course Prerequisite:			
Basic knowledge of structural mechanics & engineering mathematics.			
Course Objectives:			
1. Aware of Earthquake terminologies 2. Learn the behaviour of structures under dynamic load 3. Earthquake principles at the planning & design stage of RC structures 4. Apply Earthquake resistant features & ductility during Construction of RC structures			
Course Outcomes:			
After completing the course, students will be able to			
1. Describe the causes & effects of an Earthquake and various Earthquake terminologies 2. Perform preliminary dynamic analysis of structures under earthquake load. 3. Apply the principles of earthquake-resistant design philosophy in planning, design of RC buildings. 4. Adopt the provisions of latest IS 1893 and IS 13920 Codes. 5. Incorporate the Earthquake resistant features in RC buildings.			
Unit – I	Elements of Seismology		4 Hours
General effects of an earthquake, EQ. terminology, structure of earth, causes of an earthquake, plate tectonic theory, seismic waves, magnitude and intensity, methods of measurement, energy released, seismograph, soil liquefaction, prominent earthquakes of India			
Unit – II	Theory of Vibration		3 Hours



Dynamic loads and dynamic analysis, degrees of freedom, Multiple elastic forces-numerical problems, Types of damping relevance to buildings		
Unit – III	Free Vibration of single storied building	4 Hours
Cases of Damping, Undamped & Damped vibration, Structural Idealisation & application		
Unit – IV	Forced Vibration of single storied building	4 Hours
Forced vibrations (harmonic loading). Undamped and damped vibrations-equations of motion, Structural Idealisation & application, Concept of Force transmitted to foundation of structure		
Unit – V	Principles of Earthquake resistant design	4 Hours
Principles: Planning aspects, symmetry, simplicity, regularity. Resistance of structural elements and structures for seismic load, design criteria, strength and deflection.		
Unit – VI	Evaluation of Seismic Forces	4 Hours
Philosophy of earthquake resistant design, Provisions of latest IS 1893, Soft storey, Design spectrum of latest IS1893, evaluation of lateral loads due to earthquake on multi-storey buildings.		
Unit – VII	Ductile detailing of RCC members	4 Hours
Concept of ductility, different ways of measuring ductility, factors affecting ductility, energy absorption, provisions of latest IS 13920, Application to typical structure		
Unit – VIII	Earthquake-resistant construction	3 Hours
Repair, Retrofitting and strengthening techniques of low-rise buildings. Provisions of IS 4326 and IS 13935		
Note: Assignments based on every unit shall be solved during the practical session. IS: 1893 & 13920 latest codes are mandatory during course learning interaction.		
Text Books		
1. Elements of Earthquake Engineering –Jai Krishna,South Asian Pub. New Delhi 2. Earthquake Resistant Design of Structures–A.S. Arya 3. Earthquake Resistant Design of R.C.C. Structures–S.K.Ghosh		
Reference Books		
1. Dynamics of Structures–A.K. Chopra 2. Structural Dynamics- Mario Paz CBS Publication 3. Earthquake Resistant Structures–D.J. Dowrick John Wiely Publication 4. Dynamics of Structures–R.M. Clough and Penzian,McGraw Hill co.New Delhi		



5. Mechanical Vibrations–G.R. Grover, Roorkee University, Roorkee
6. Analysis and Design of Foundations for Vibrations – P.J.Moove. Oxford and I.B. H. Publication, Delhi
7. Foundation Design Manual–N.V.Nayak, Dhanpat rai and sons, Delhi
8. Manual of Earthquake Resistant Non-Engineering Construction, University of Roorkee
9. Elements of Seismology– Rochter
10. Latest IS 1893 &13920, IS: 4326 and IS 13935.
11. Earthquake Tips published by NICEE, IIT Kanpur.

e-Resources

1. www.nicee.org
2. EFNARC Global specifications





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VII

23CEU7E24T: Program Elective Course V-Urban Transportation Planning

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

Urban Transportation Planning focuses on the planning and analysis of transportation systems in urban areas to address issues such as congestion, safety, accessibility, and environmental sustainability. The course introduces students to urban transport problems, policies, travel characteristics, and planning frameworks, along with methods of data collection and travel demand modeling. Emphasis is placed on the planning of urban transport networks, corridor identification, and preparation of urban transportation plans in line with contemporary practices such as the National Urban Transport Policy and sustainable mobility principles.

Course Prerequisite:

Basic understanding of transportation engineering fundamentals and Traffic Engineering.

Course Objectives:

1. Provide an understanding of urban transportation problems, policies, and planning principles in the context of rapidly growing cities.
2. Introduce students to urban travel characteristics and transportation system approaches involving supply and demand interactions.
3. Develop competence in transportation data collection, analysis, and inventory preparation for planning studies.
4. Familiarize students with travel demand modeling techniques, including trip generation, trip distribution, mode split, and traffic assignment.
5. Enable students to plan urban transport networks, identify corridors, and prepare comprehensive urban transportation plans considering sustainability and multimodal integration.

Course Outcomes:

After completing the course, students will be able to

1. Identify and analyze urban transportation problems and policy issues using a systems approach.
2. Develop and organize transportation databases through appropriate surveys and data collection



techniques for urban planning studies. - Estimate trip generation and trip distribution using zonal, growth factor, and gravity-based models. - Analyze mode choice behavior and apply suitable mode split models for urban travel demand estimation - Perform traffic assignment on urban transport networks using capacity restraint and equilibrium assignment techniques.		
Unit – I	Urban Transportation Problems and Policy	5 Hours
Urban transportation Issues, Travel Characteristics, Evolution of Planning Process, Supply and Demand – Systems approach; NUTP, Recommendations of 12th FYP and NTDP; Smart Cities, Service Level Benchmarks.		
Unit – II	Travel Demand Approaches	5 Hours
Trends, Overall Planning process, long term Vs. Short-term planning, Types of Plans, Master Plans, Demand Function, Independent Variables, Travel Attributes, Assumptions in Demand Estimation, Sequential, and Simultaneous Approaches, Aggregate and Disaggregate Techniques, UTPS Approach.		
Unit – III	Data Collection and Inventories	6 Hours
Collection of data – Organisation of surveys and Analysis, Study Area, Zoning, Types, and Sources of Data, Road Side Interviews, Home Interview Surveys, Commercial Vehicle Surveys, Sampling Techniques, Expansion Factors, Accuracy Checks, Use of Secondary Sources, Economic data – Income – Population – Employment – Vehicle Owner Ship.		
Unit – IV	Trip Generation and Trip Distribution	6 Hours
Trip Generation Analysis: Zonal Models, Category Analysis, Household Models, Trip Attraction models, Commercial Trip Rates. Trip Distribution: Growth Factor Methods, Gravity Models, Opportunity Models, Time Function Iteration Models.		
Unit – V	Mode Split and Traffic Assignment	10 Hours
Mode Choice Behaviour, Competing Modes, Mode Split Curves, Models, and Probabilistic Approaches. Traffic Assignment: Basic Elements of Transport Networks, Coding, Route Properties, Path Building Criteria, Skimming Tree, All-or-Nothing Assignment, Capacity Restraint Techniques, Reallocation of Assigned Volumes, Equilibrium Assignment, Diversion Curves.		
INTERNAL CONTINUOUS ASSESSMENT (ICA) Internal Continuous Assessment (ICA) shall consist of a minimum of six assignments based on the entire curriculum, which will be assessed by faculty.		
Text Books		
Transportation Engineering and Planning; Papacostas, C.S. and Prevedouros, P.D., Prentice Hall of India Pvt. Ltd., 2012, Third Edition.		



2. Transportation Engineering, Khisty C.J. and B Kent Lall; Prentice Hall of India Pvt. Ltd., 2012, Third Edition.
3. Urban Transportation Planning Lecture Notes, Chari, S.R., NIT Warangal, 1990.

Reference Books

1. Modelling, Transport, Juan de Dios Ortúzar and Luis G. Willumsen, *Modelling Transport*, **3rd Edition**, John Wiley & Sons, New York, USA.
2. Principles of Urban Transportation System Planning, B. G. Hutchinson, *Principles of Urban Transportation System Planning*, McGraw Hill, New York, USA.
3. Urban Transportation Planning: A Decision-Oriented Approach, Michael D. Meyer and Eric J. Miller, *Urban Transportation Planning: A Decision-Oriented Approach*, **2nd Edition**, McGraw Hill, New York, USA.
4. Transportation Planning Handbook, John D. Edwards (Ed.), *Transportation Planning Handbook*, **2nd Edition**, Institute of Transportation Engineers (ITE), Washington D.C., USA.
5. An Introduction to Transportation Planning, Michael J. Bruton, *An Introduction to Transportation Planning*, UCL Press, London, UK.

e-Resources

1. **NPTEL**, *Urban Transportation Planning* – Video lectures and notes by IIT faculty.
2. **MIT OpenCourseWare**, *Urban Transportation Planning* (Course materials, assignments, and lecture notes).
3. **World Bank Open Learning Campus**, *Integrated Urban Transport Planning* – Self-paced learning modules.
4. **Coursera**, Courses on *Urban Mobility and Transportation Planning* by international universities.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
Final Year B.Tech. (Civil Engineering), Semester - VII

23CEU7E34T: Program Elective Course V-Environmental Audits & Industrial Compliance

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Introduction:

This course provides a comprehensive understanding of environmental auditing frameworks, statutory requirements, and compliance mechanisms as per the Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs). It covers various types of environmental audits, assessment methodologies, reporting, and corrective action planning for sustainable industrial operations.

Course Prerequisite:

Basic knowledge of Environmental Engineering and Environmental Laws

Course Objectives:

1. To familiarize students with CPCB guidelines and legal frameworks for environmental audits.
2. To develop skills in planning, conducting, and reporting environmental audits in industries.
3. To enable students to identify non-compliance issues and recommend corrective measures.
4. To expose students to emerging trends in environmental compliance and sustainability reporting.

Course Outcomes:

After completing the course, students will be able to

1. Interpret CPCB/SPCB guidelines and statutory requirements for various industries.
2. Plan and conduct different types of environmental audits (water, air, waste, energy).
3. Analyze audit findings and prepare compliance and non-compliance reports.
4. Recommend mitigation strategies and environmental management plans for industries.

Unit – I	Introduction to Environmental Auditing	6 Hours
Definition, scope, and importance; Types of audits: Compliance, Performance, Waste, Energy, Water, EIA, follow-up; Legal framework: CPCB, SPCBs, MoEF & CC; Overview of Environmental Protection Act, 1986, Water & Air Acts, Hazardous Waste Rules, EIA Notification.		
Unit – II	Audit Methodologies and Tools	8 Hours



Audit planning, pre-audit activities, checklists, questionnaires; Data collection: sampling, monitoring, records review; On-site inspection techniques; Use of IoT and remote sensing in auditing.		
Unit – III	Water Audit and Wastewater Compliance	8 Hours
CPCB guidelines for water conservation and wastewater discharge; Water balance, source identification, flow measurement; Effluent standards, CETP/ETP compliance; Case studies of industrial water audits.		
Unit – IV	Air Quality and Emissions Audit	6 Hours
Stack monitoring, ambient air quality assessment; CPCB norms for particulate and gaseous emissions; Fugitive emissions, odour management; Audit of pollution control equipment (ESP, scrubbers, etc.).		
Unit – V	Waste and Resource Audit	6 Hours
Solid and hazardous waste audit as per CPCB; Waste minimization, recycling, and recovery; Energy audit and carbon footprint assessment; Circular economy concepts in industries.		
Unit – VI	Audit Reporting and Environmental Management Plan	6 Hours
Structure of audit report, non-compliance documentation, corrective and preventive actions; Environmental Management System (ISO 14001), EMS auditing; Sustainability reporting (GRI, BRSR).		
Six assignments covering each unit, including: • Review of CPCB guidelines for a specific sector • Preparation of partial audit reports • Compliance gap analysis for a given scenario		
Text Books		
1. CPCB Publications: "Guidelines for Environmental Audit", "Water Audit Manual", "Emission Standards". 2. Modi, P. N., "Environmental Management and Audit", Standard Book House, 2020. 3. Kumar, E. S., "Environmental Auditing and Compliance", PHI Learning, 2019.		
Reference Books		
1. ISO 14001:2015 Environmental Management Systems – Requirements. 2. MoEF&CC Guidelines on EIA and Environmental Clearance. 3. "Industrial Environmental Management" by R. K. Trivedi, 2021.		
e-Resources		



1. CPCB Official Website – Guidelines and Reports
2. NPTEL Course: Environmental Auditing and Life Cycle Assessment
3. MoEF & CC E-Resource Portal





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B. Tech. (Civil Engineering), Semester-VII

23CEU7CC6L: RCC Structural Design and Drawing

Teaching Scheme		Examination Scheme	
Practical	2 Hours/week	ICA	25 Marks
Credits	1	POE	25 Marks
Introduction:			
This course covers a complete 3D model of a multi-storey RC building for G+5 and analysing it for gravity and seismic loads using commercial structural engineering software. The course also covers the design of one specialised RC component, such as a combined footing, raft foundation, pile foundation with pile cap, or water tank using codal guidelines.			
Course Prerequisite:			
Understanding of basic structural analysis principles, including static and dynamic analysis of structures. Familiarity with Indian standard codes. A background in reinforced concrete design using limit state and working stress methods is essential. A good grasp of engineering mechanics and strength of materials will support better judgment during design and detailing.			
Course Objectives:			
1. Develop the ability to model and analyse a multi-storey RC framed building in 3D using commercial structural software under gravity & seismic load combinations as per IS code provisions. 2. Analyse the building model and interpret analysis results for the design and detailing of RC structural elements as per relevant codes and prepare structural design and detailed drawings. 3. Develop the ability to design selected specialized RC components using appropriate codal provisions and design methods.			
Course Outcomes:			
After completing the course, students will be able to			
1. Analyze 3D multi-storey RC framed buildings by applying IS code load combinations, including gravity and earthquake loads using structural software. 2. Evaluate and interpret design forces from software outputs to design slabs, beams, columns, and footings in compliance with codal provisions and prepare detailed structural drawings. 3. Analyse and design for selected specialized RC components and prepare full structural detailed			



drawings.		
Unit – I	3D Analysis and Design of RC Building for G + 5	9 Hours
3D modelling and analysis of an RC-framed building structure under design load combinations, including gravity and earthquake loads. Use of commercial software. Evaluate design forces from software and design of structural elements using professional/open-source software or self-developed computational tools		
Unit – II	Anyone from the following	5 Hours
a) Combined trapezoidal footing/ raft foundation. b) Pile foundation for the structure with a pile cap. c) Water tank (GSR/USR/ESR) by working stress method using IS 3370. Analysis and design of specialised RCC components: load assessment, structural behaviour, proportioning, and reinforcement detailing in accordance with relevant Indian Standards; prepare complete design calculations and structural drawings for the selected system.		
Note: a) Sample verification of software analysis results shall be validated using analytical methods. b) Maximum number of students in a group not more than five for design work.		
Internal Continuous Assessment (ICA) 1. Carry out 3D Computer analysis and full design of RCC building and prepare design report. 2. Structural drawing showing reinforcement detailing in A2 size drawing sheet. 3. Site visits: The report should contain structural details with sketches.		
Text Books		
1. Reinforced Concrete Design, by S.U. Pillai and D. Menon, Tata McGraw Hill, 3rd Edition. 2. Earthquake Resistant Design of Structures, by P. Agarwal and M. Shrikhande, Prentice-Hall of India Private Limited, 2006. 3. Limit State Design of Reinforced Concrete, by-P.C. Varghese, Prentice Hall of India.		
Reference Books		
1. Seismic Design of Reinforced Concrete and Masonry Buildings, by T. Paulay and M.J.N. Priestley, John Wiley and Sons Inc., 1992. 2. Earthquake Resistant Design of Structures, by S.K. Duggal, Oxford University Press, 2007.		
Indian Standard Codes		
1. IS 456:2000, Plain and Reinforced Concrete - Code of Practice, Bureau of Indian Standards,		



New Delhi.

2. IS 1893 (Part 1): 2016, Criteria for Earthquake Resistant Design of Structures, Bureau of Indian Standards, New Delhi.
3. IS 1893(Part 2): 2014 Criteria for Earthquake Resistant Design of Structures (P-II) Liquid Retaining Tanks, Bureau of Indian Standards, New Delhi.
4. IS 13920:2016, Code of Practice for Ductile Detailing of Reinforced Concrete Structures subjected to Seismic Forces, Bureau of Indian Standards, New Delhi.
5. IS 3370 (Part I): 2021, Code of Practice for Concrete Structures for Retaining Aqueous Liquids. Part I General Requirements. Bureau of Indian Standards, New Delhi.
6. IS 3370 (Part II): 2021, Code of Practice for Concrete Structures for Retaining Aqueous Liquids. Part II Plain and Reinforced Concrete Structures. Bureau of Indian Standards, New Delhi.
7. IS 3370 (Part IV): Latest, Code of Practice for Concrete Structures for Retaining Aqueous Liquids. Part IV Design Tables. Sec 3 Circular Tanks Bureau of Indian Standards, New Delhi.
8. IS 11682 (1985): Criteria for Design of RCC Staging for Overhead Water Tanks.

e-Resources

1. Web links and Video Lectures (e-Resources):
 - a) <https://www.youtube.com/playlist?list=PL51300B0778FB5784>
 - b) <https://www.youtube.com/watch?v=AJG1Jix8NDk>
 - c) <https://www.civilconcept.com/online-tools-to-design-two-way-slab/>
 - d) https://www.slideshare.net/slideshow/design-of-reinforced-concrete-structure-is-4562000-104570451/104570451?from_search=2
 - e) <https://www.slideshare.net/slideshow/gate-rcc/250568427>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)

Final Year B.Tech. (Civil Engineering), Semester - VII

23CEU7PR7L: Project Phase-I

Teaching Scheme		Examination Scheme	
Practical	8 Hours/week	ICA	100 Marks
Credits	4	OE	50 Marks

Introduction:

Project-based learning is a paradigm that is becoming time-honoured nowadays. To keep abreast with this, the Project course is included in the curriculum, which is spread over both semesters of the final year. For this course, students carry out a project as a team that allows them to demonstrate their abilities and develop skills within their chosen area of interest. Any subdomain of Civil Engineering and Multidisciplinary with Civil Engineering relevance projects with a focus on design and research aspects are accepted. Also, communicating effectively, both in oral and written form is an important skill for engineering graduates in many different contexts. This course also aims to foster these skills.

Course Prerequisite:

Students shall have technical competency as well as behavioral facets to carry out projects as a part of a team. The student shall have an adept knowledge of core Civil Engineering subjects with software competence. Students shall also possess the necessary technical report writing and, presentation skills and shall have proficiency in office software for word processing and presentation.

Course Objectives:

1. To expose students to the application of civil engineering and allied fields for societal benefit and to identify the problem area and formulate the problem statement.
2. To make students aware of the present trends used to solve the problem undertaken
3. To make students able to adopt mathematical/automation/analysis/design according to the scope of the problem.
4. To enable students to conduct preliminary experiments, and simulations to solve the problem undertaken.
5. To make student able to write technical specifications, a n d project documents over problems undertaken.

Course Outcomes:



After completing the course, students will be able to

1. Identify the problem area and formulate the problem statement
2. Analyze the present trends used to solve the problem
3. Design the system according to the scope of the problem
4. Conduct preliminary experiments, simulations, and modeling techniques to solve the problem
5. Write a synopsis, technical specification, and project report on the solution to the problem undertaken.

The objective of **Project Phase – I** is to enable the student to take up investigative study in the broad field of Civil Engineering and allied fields, either fully practical or involving both theoretical and practical work to be assigned by the department on the basis of five students in a group, under the guidance of a supervisor.

This is expected to provide a good initiation for the students in further career building. The tasks during Project Phase – I include:

1. Survey and study of published literature on the assigned topic.
2. Working out a preliminary Approach to the Problem relating to the assigned topic.
3. Conducting preliminary Analysis / Modeling / Simulation / Experiment /Design / Feasibility.
4. Preparing a written report on the study conducted for presentation to the department.
5. Seminar as an oral presentation in front of a departmental assessment & evaluation committee.



**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B. Tech. (All Branches), Semester-VIII****23CMU8R11T: Research Methodology and Intellectual Property Rights****Teaching Scheme****Lectures**

-

Credits

2

Examination Scheme**ESE**

50 Marks

Introduction

Research Methodology provides a systematic framework for identifying, analyzing, and solving technical problems through evidence-based inquiry. Intellectual Property Rights (IPR) establish the legal safeguards necessary to protect these engineering innovations, ensuring that creators maintain control and commercial benefits over their inventions. Together, they equip students with the tools to conduct ethical research and navigate the global patent and copyright landscape.

Course Prerequisites:

Basics of Engineering and Statistics

Course Objectives:***This course aims to:***

1. Impart knowledge on research types, processes, design, and literature review techniques.
2. Develop skills in data collection, sampling, and statistical analysis of research data.
3. Enable preparation of research reports with standard citations and inculcate research ethics.
4. Provide understanding of IPR types, legal frameworks, patent search, and filing process.

Course Outcomes:***At the end of this course, students will be able to:***

1. Explain the fundamental concepts of research design, its types, processes, and systematic literature review techniques.
2. Analyze research data by using appropriate data collection methods, sampling techniques, and statistical tools.
3. Prepare research reports using standard citation formats and demonstrate ethical responsibilities in research conduct.
4. Classify IPR types and legal frameworks, and apply knowledge to patent search and filing process.

Unit – I**Introduction to Research**

Meaning & objectives of research, Types of research: basic, applied, quantitative, qualitative, Steps in research process, identifying research problems, Formulating hypotheses and objectives		
Unit – II	Literature Review & Research Design	
Literature search strategies, IEEE Xplore, Scopus, Google Scholar, Springer, Web of Science; Research gap analysis, Research design: exploratory, descriptive, experimental, Sampling techniques		
Unit – III	Data Collection & Analysis	
Primary vs secondary data, Tools of data collection (questionnaires, surveys, interviews), Basics of statistical analysis: Mean, Median, Mode, Standard deviation, correlation; Use of software (Excel, SPSS), Data interpretation and presentation.		
Unit – IV	Technical Communication & Ethics in Research	
Technical writing framework, Research paper structure: Abstract, Introduction, Methodology, Results, Discussion, Conclusion; Referencing and citations (APA/IEEE formats), Ethical considerations & plagiarism, Use of plagiarism checking tools		
Unit – V	Introduction to Intellectual Property Rights	
IPR definitions & relevance to engineering, Patents: types, patentable subject matter; Copyright: scope, ownership, duration; Trademarks, Trade Secrets, Industrial Designs		
Unit – VI	Patent Rules, Filing, and IP Management	
Patent search strategies, Patent drafting basics, Patent filing procedure in India and abroad, Patent Cooperation Treaty (PCT) overview, IPR protection strategy for products and software, Technology transfer, Licensing		
Text Books		
1) Research Methodology, A Practical and Scientific Approach - Vinayak Bairagi, Mousami V. Munot, Chapman and Hall 2) Research Methodology: C.R. Kothari , New Age International Publications 3) Research methodology: A step-by-step guide for beginners - Kumar, R. (2018) 4) Intellectual Property Rights: P. Narayanan, S. Chand Publication		
Reference Books		
1) “Research Methodology: Methods & Techniques” — Ranjit Kumar, Sage Publications 2) Intellectual Property Rights under WTO, T. Ramappa 3) Intellectual Property Handbook, WIPO (World Intellectual Property Organization, free PDF). : https://tind.wipo.int/record/28661/files/wipo_pub_489.pdf		



Relevant Videos on YouTube related to previous NPTEL Courses
1) Research Methodology — NPTEL course (Videos Available on Youtube) https://www.youtube.com/watch?v=E2gGF1rburw&list=PLyqSpQzTE6M8F_P8lgjvmqiDEoFGLzG4h&index=1
2) Intellectual Property – NPTEL course (Videos Available on Youtube) https://www.youtube.com/watch?v=a-zBaQ7Al2s&list=PLyqSpQzTE6M8PuzP1p2hNPXgpbOBhFgja
Important Websites / E-Resources
1) IEEE Xplore Digital Library: https://india.ieee.org/ieee-xplore-digital-library/
2) Google Scholar: https://scholar.google.com/
Patent Databases (url)
1) WIPO (World Int'l Patents): https://www.wipo.int/en/web/patents
2) Indian Patent Advanced Search System (InPASS): https://iprarch.ipindia.gov.in/publicsearch





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VIII

23CEU8E12T: Program Elective Course VI- Geosynthetics and Soil Structures

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Credits	2		

Introduction:

This subject introduces the principles and applications of reinforced soil and geosynthetics in modern geotechnical engineering. It focuses on the behavior, testing, design, and construction of reinforced soil structures such as retaining walls, slopes, foundations, and pavements. The course emphasizes analytical design methods, codal practices, and practical applications for safe and economical infrastructure development.

Course Prerequisite:

Basic knowledge of Geotechnical engineering and Foundation engineering.

Course Objectives:

1. Understand the fundamentals of reinforced soil systems, including historical development, soil–reinforcement interaction mechanisms, and stress behavior using Mohr circle concepts.
2. Identify and classify various geosynthetics, their manufacturing processes, and material properties for appropriate selection in geotechnical applications.
3. To develop the ability to analyze and design reinforced soil retaining walls and slopes.
4. To explain the application of geosynthetics in foundations, drainage, filtration, and erosion control.
5. To provide knowledge on the use of geosynthetics in pavement, railway, and transportation infrastructure systems.

Course Outcomes:

After completing the course, students will be able to

1. Identify the different types of geotextiles and their suitability for soil reinforcement structures.
2. Classify different types of geosynthetics and select suitable materials based on functional requirements such as separation, reinforcement, filtration, and drainage.
3. Analyze and design reinforced soil retaining walls and slopes using codal methods and stability analysis techniques.
4. Design reinforced soil retaining walls and slopes using established methods (BS 8006, FHWA) and perform stability analyses using wedge and circular slip methods.
5. To apply geosynthetics in foundations, pavements, erosion control, drainage, and railways using analytical design methods.



Unit – I	Fundamentals of Reinforced Soil and Geosynthetics
Introduction: Historical background of reinforced soil, Principles of reinforced soil through Mohr circle analysis. Different types of geosynthetics: Types of geosynthetics like geotextiles, geogrids, geonets, geocells, geo-composites, and their manufacturing methods.	
Unit – II	Testing and Characterization of Geosynthetics
Testing methods for geosynthetics: Techniques for testing of different index properties, strength properties, Apparent Opening Size, In-plane and cross-plane permeability tests, assessment of construction induced damage, and extrapolation of long-term strength properties from short term tests.	
Unit – III	Reinforced Soil Retaining Walls and Slopes
Reinforced Soil retaining walls: Different types of walls like wrap-around walls, full-height panel walls, discrete-facing panel walls, and modular block walls. Design methods as per BS-8006 and FHWA methods, Construction methods for reinforced soil retaining walls. Reinforced soil slopes: Basal reinforcement for construction on soft clay soils, construction of steep slopes with reinforcement layers on competent soils, Different slope stability analysis methods like planar wedge method, bi-linear wedge method, and circular slip methods.	
Unit – IV	Foundation, Drainage, and Erosion Applications
Erosion control on slopes using geosynthetics. Applications in foundations: Binquet and Lee's approach for analysis of foundations with reinforcement layers. Drainage and filtration applications of geosynthetics: Different filtration requirements, filtration in different types of soils, and criteria for selection of geotextiles, estimation of flow of water in retaining walls, pavements, etc., and selection of geosynthetics.	
Unit – V	Transportation and Pavement Applications of Geosynthetics
Pavement application: Geosynthetics for separation and reinforcement in flexible pavements, designed by the Giroud-Noiray approach, reflection cracking, and control using geosynthetics. Use of geosynthetics for the construction of heavy container yards and railway lines.	
Text Books	
1. Koerner, R.M. "Designing with Geosynthetics", Prentice Hall, New Jersey, USA, 4th edition, 1999. 2. Jewell, R.A., "Soil Reinforcement with Geotextiles", Special Publication No. 123, CIRIA, Thomas Telford. London, UK, 1996. 3. Geosynthetics - New Horizons, Eds. G.V. Rao, PK Banerjee, J.T. Shahu, G.V. Ramana, Asian Books Private Ltd., New Delhi, 2004.	
Reference Books	
1. Geosynthetics Asia 1997: Select papers by C.V.J. Varma, G. Venkatappa Rao and A.R.G. Rao, 1998. 2. Geosynthetics for Trails in Wet Areas: 2008, Edition by James Scott Groenier, Bibliogov Publishers, 2012.	



3. Fundamentals of Geosynthetic Engineering by Sanjay Kumar Shukla and Jian-Hua Yin, Taylor & Francis, 2008.

e-Resources

1. Web links and Video Lectures (e-Resources):

<https://nptel.ac.in/courses/105101143>





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VIII

23CEU8E22T: Program Elective Course VI- Remote Sensing and GIS

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Credits	2		

Introduction:

Remote Sensing and Geographic Information System (GIS) are powerful tools for acquiring, analyzing, and managing spatial and attribute data related to the Earth's surface. This course introduces the fundamental principles of remote sensing, satellite platforms and sensors, electromagnetic energy interactions, and digital image processing techniques. It also covers GIS concepts, spatial data models, coordinate systems, and integrated applications of Remote Sensing and GIS in civil engineering, urban planning, transportation, water resources, and environmental management.

Course Prerequisite:

Require Basic knowledge of Engineering Mathematics, Surveying, Environmental Studies, and Computer fundamentals

Course Objectives:

1. Introduce the fundamental concepts and principles of Remote Sensing and GIS.
2. Explain the electromagnetic spectrum and energy interactions with Earth surface features.
3. Familiarize students with satellite platforms, sensors, and digital image processing techniques.
4. Develop understanding of GIS data models, coordinate systems, and map projections.
5. Demonstrate integrated applications of Remote Sensing and GIS in civil engineering and infrastructure planning.

Course Outcomes:

After completing the course, students will be able to,

1. Explain basic principles and processes of Remote Sensing.
2. Identify remote sensing platforms and sensors such as IRS, Landsat, and Cartosat with their resolutions.
3. Apply basic digital image processing and visual interpretation techniques.



4. Describe GIS concepts, spatial data models, coordinate systems, and map projections. 5. Apply Remote Sensing and GIS for land use, urban, environmental, and transportation applications.	
Unit – I	Remote Sensing
Basic concept of Remote sensing, Data and Information, Remote sensing data collection, Remote sensing advantages & Limitations, Remote Sensing process. Electromagnetic Spectrum, Energy interactions with atmosphere and with earth surface features (soil, water, and vegetation), Resolution, image registration and Image and False color composite, elements of visual interpretation techniques.	
Unit – II	Remote Sensing Platforms and Sensors
Indian Satellites and Sensors characteristics, Remote Sensing Platforms, Sensors and Properties of Digital Data, Data Formats: Introduction, platforms- IRS, Landsat, SPOT, Cartosat, Ikonos, Envisat, etc., sensors, sensor resolutions (spatial, spectral, radiometric and temporal). Basics of digital image processing- introduction to digital data, systematic errors (Scan Skew, Mirror-Scan Velocity, Panoramic Distortion, Platform Velocity, Earth Rotation) and non-systematic [random] errors (Altitude, Attitude), Image enhancements (Grey Level Thresholding, level slicing, contrast stretching), image filtering.	
Unit – III	Geographic Information System
Introduction to GIS; components of a GIS; Geographically Referenced Data, Spatial Data- Attribute data, Joining Spatial and attribute data, GIS Operations: Spatial Data Input – Attribute data Management, Geographic coordinate System, Datum; Map Projections: Types of Map Projections, Projected coordinate Systems. UTM Zones.	
Unit – IV	Data Models
Vector data model: Representation of simple features – Topology and its importance; coverage and its data structure, Shape file; Relational Database, Raster Data Model: Elements of the Raster data model, Types of Raster Data, Raster Data Structure, Data conversion.	
Unit – V	Integrated Applications of Remote Sensing and GIS



Applications in land use and land cover analysis, change detection, water resources, urban planning, environmental planning, Natural resource management and Traffic management. Location-Based Services and its Applications.

Text Books

1. Lillesand, T. M., Kiefer, R. W., & Chipman, J. W. – *Remote Sensing and Image Interpretation*, Wiley.
2. Burrough, P. A. & McDonnell, R. A. – *Principles of Geographical Information Systems*, Oxford University Press.
3. Bhatta, B. – *Remote Sensing and GIS*, Oxford University Press

Reference Books

1. Campbell, J. B. & Wynne, R. H. – *Introduction to Remote Sensing*, Guilford Press.
2. DeMers, M. N. – *Fundamentals of Geographic Information Systems*, Wiley.
3. Jensen, J. R. – *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson.
Lo, C. P. & Yeung, A. K. W. – *Concepts and Techniques of Geographic Information Systems*, Prentice Hall.

e-Resources

1. ISRO – IIRS Outreach Programme: Online lectures and tutorials on Remote Sensing and GIS
2. QGIS Documentation and Tutorials
3. ArcGIS Online Learning Platform (ESRI Training)
4. NPTEL Courses:
Remote Sensing
Geographic Information Systems





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VIII

23CEU8E32T: Program Elective VI- Planning and Design of Rural Roads

Teaching Scheme		Examination Scheme	
Lectures	-	ESE	50 Marks
Credits	2		

Introduction:

Rural roads play a vital role in improving connectivity, socio-economic development, and access to essential services in rural areas. This course provides comprehensive knowledge of planning, geometric design, pavement design, drainage, construction, quality control, and maintenance of rural roads. Emphasis is given to national guidelines and practices adopted in India, including rural road development programs such as PMGSY and standards prescribed by IRC. The course equips students with practical skills required for sustainable, economical, and safe rural road infrastructure development.

Course Prerequisite:

Require Basic knowledge of **Surveying**, Introductory concepts of **Soil Mechanics / Geotechnical Engineering**, and Basic understanding of **Transportation / Highway Engineering**

Course Objectives:

1. Introduce the principles of rural road planning, network development, and route alignment.
2. Familiarize students with geometric design standards for rural roads.
3. Develop competence in soil and material investigations and pavement design for rural roads.
4. Explain drainage requirements, culvert and small bridge design for rural roads.
5. Impart knowledge of construction practices, specifications, quality control, and maintenance of rural roads.

Course Outcomes:

After completing the course, students will be able to,

1. Understand rural road planning concepts, network planning, alignment selection, and guidelines, including PMGSY.
2. Apply geometric design standards for rural roads as per terrain, traffic, and safety requirements.
3. Analyse soil and material survey data and design flexible, rigid, semi-rigid, and special



pavements as per IRC guidelines. 4. Design suitable road drainage systems, culverts, and small bridges for rural roads. 5. Understand construction methods, specifications, quality control, and maintenance practices for various types of rural road pavements.	
Unit – I	Planning of Rural Roads
Planning database, the concept of network planning, Rural roads plan, guidelines laid down in recent 20-year plans and in PMGSY Road alignment and surveys, governing factors for route selection Factors controlling alignment; obligatory points, traffic, geometric designs, economy, special considerations in hilly areas.	
Unit – II	Geometric Design Standards
Classification of rural roads, terrain classification, design speed, basic principles of geometric design, cross-sectional elements, camber, and sight distances. Horizontal alignment: general guidelines, superelevation, transition curve, widening and setback distances, vertical alignment: gradient, grade compensation at curves, valley curves, alignment compatibility, lateral and vertical clearances.	
Unit – III	Soil and Material Surveys
Soil investigations for low embankment, high embankment, cut sections, subgrade, Survey for marginal materials and aggregates/low-grade materials Artificial aggregates, waste materials, new materials, and stabilizers Design parameters, pavement components Design of flexible pavement as per IRC: pavement thickness, pavement surfacing Design of semi-rigid pavement: dry lean concrete/lime fly ash concrete bases Design of rigid pavement: cement concrete pavement Design of special pavements: concrete block pavement , interlocking concrete block pavement Choice of pavement type and material.	
Unit – IV	Types of Road Drainage
Requirements of surface drain; roadside drains, shoulder drains, catch water drains. Requirements subsurface drain. Cross drains; types, requirements, and choice of different types of cross drain. Standard designs of culverts. The standard design of small bridges.	
Unit – V	Construction and Specification
Construction and Specifications: Earth, Sub-base, Base course, and surface course – materials, specifications, and construction steps and use of different equipment, construction of special pavements, construction of fly ash embankments, lime fly ash stabilized soil, lime fly ash bound macadam, lime fly ash concrete, roller compacted concrete, dry lean fly ash concrete, cement stabilized fly ash, quality control in construction, Specifications, and tests for quality control as per IRC.	
Unit – VI	Maintenance of Rural Roads



Types of maintenance: maintenance of unpaved roads, maintenance of paved roads, maintenance of semi-rigid and roller compacted concrete pavements, maintenance of special pavements, and rehabilitation.

Text Books

1. Khanna, S.K., Justo, C.E.G., and Veeraragavan, A., 'Highway Materials and Pavement Testing', Nem Chand and Bros, Roorkee Reference Books:
2. IRC SP 20 Rural Roads Manual
3. Ministry of Rural Road Development

Reference Books

1. Principles of Transportation Engineering, Partha Chakroborty and Animesh Das, PHI Publication.
2. Transportation Engineering – An Introduction, Khistry, C.J., PHI Publication.
3. Specifications of Road and Bridge Works (MoRTH) Publication – 5th Revision. New Delhi.
4. IRC: 37-2018, IRC: 58-2015, and other relevant IRC codes

e-Resources

1. **NPTEL:** Courses on *Highway Engineering and Pavement Design*
2. **IRC:** IRC guidelines, codes, and special publications
3. **MoRTH:** Specifications and standards for road works
4. **PMGSY:** Rural road manuals, technical guidelines, and case studies
5. **Virtual Labs (MHRD/AICTE):** Pavement materials and highway engineering experiments





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VIII

23CEU8IN3L: Internship

Teaching Scheme		Examination Scheme	
Practical	20 Hours/week	ICA	100 Marks
Credits	10	OE	50 Marks

Introduction:

The Internship course provides Civil Engineering students with practical exposure to real-world engineering practices in industries, construction sites, consultancy firms, and infrastructure projects. It helps students apply theoretical knowledge to practical problems related to planning, design, construction, quality control, surveying, and project management. The internship also develops professional skills such as teamwork, communication, ethics, and technical reporting required for a successful civil engineering career.

Course Prerequisite:

Students must have completed the prerequisite courses of the Civil Engineering program, including core subjects related to Structural Engineering, Geotechnical Engineering, Transportation Engineering, Environmental Engineering, Surveying, Construction Technology, and Project Management. Students should possess basic knowledge of civil engineering practices such as preparation and interpretation of engineering drawings, estimation and costing, material testing procedures, site surveying, and safety regulations followed in construction projects.

Course Objectives:

1. To provide students with practical exposure to real-world Civil Engineering activities such as planning, design, construction, surveying, quality control, estimation, and project management.
2. To enable students to apply theoretical knowledge gained in Civil Engineering subjects to practical problems encountered in construction sites, consultancy services, laboratories, and infrastructure projects.
3. To develop technical and professional skills required in Civil Engineering industries, including use of engineering tools, preparation of reports, teamwork, communication, and problem-solving abilities.
4. To familiarize students with professional practices, safety standards, ethical responsibilities, and workplace discipline followed in Civil Engineering organizations.



5. To provide students with industry-oriented learning experiences that are not available in the classroom environment and enhance their employability in the Civil Engineering sector.
6. To encourage students to explore various career opportunities in Civil Engineering fields such as structural engineering, transportation engineering, geotechnical engineering, environmental engineering, construction management, and infrastructure development.

Course Outcomes:

After completion of the internship, students will be able to:

1. Apply the knowledge of Civil Engineering principles to solve real-world problems related to construction, surveying, transportation, structural, geotechnical, and environmental engineering projects.
2. Demonstrate teamwork, coordination, and leadership qualities while working on Civil Engineering site activities and project assignments.
3. Interact effectively with engineers, contractors, consultants, and other industry professionals while understanding professional ethics and workplace responsibilities in Civil Engineering practice.
4. Identify and demonstrate professional behavior, safety practices, discipline, and ethical conduct expected in Civil Engineering organisations and construction environments.
5. Recognize and describe Civil Engineering site practices, technical terminology, construction methods, industry trends, and challenges associated with infrastructure and development projects.

Assessment method:

The student has to complete a semester-long internship of a minimum of two months duration at industry during Final Year Sem VIII. Work done by the student in the industry and the related report will be accepted as an internship Report'.

After completion of the internship, the student should prepare a comprehensive report that represents the learning during the internship period. The student may contact, if required, his Industrial Supervisor / HOD / TPO for assigning special topics and problems, and should prepare the final report on the assigned topics. The training report should be signed by the Internship Supervisor and HOD.

The student shall present a report based on their internship before an expert committee constituted by the relevant department, as per the institute's norms.

The evaluation will be based on the following criteria:

- Proper presentation skills.
- Effectiveness of presentation.
- Depth of knowledge and skills.
- Content of internship report.
- External Oral Examination.





WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Civil Engineering), Semester-VIII

23CEU8PR4L: Project Phase II

Teaching Scheme

Practical 4 Hours/week

Credits 2

Examination Scheme

ICA 100 Marks

OE 50 Marks

Introduction:

Project-based learning is a paradigm that is becoming time-honoured nowadays. To keep abreast with this, the Project course is included in the curriculum, which is spread over both semesters of the final year. For this course, students carry out a project as a team that allows them to demonstrate their abilities and develop skills within their chosen area of interest. Any subdomain of Civil Engineering and Multidisciplinary with Civil Engineering relevance projects with a focus on design and research aspects are accepted. Also, communicating effectively, both in oral and written form, is an important skill for engineering graduates in many different contexts.

Course Prerequisite:

Students shall have technical competency as well as behavioural facets to carry out projects as a part of a team. He shall have an adept knowledge of core Civil Engineering subjects with software competence. Students shall also possess the necessary technical report writing and presentation skills, and shall have proficiency in office software for word processing and presentation.

Course Objectives:

1. To make students apply design concepts, prepare a detailed planning to solve the problem undertaken
2. To make students evaluate and analyse the performance of the proposed solution to the problem undertaken
3. To make the student aware of his responsibilities, working in a team to provide time-bound solutions to the problem
4. To make students write technical specifications and project documents on problems undertaken.
5. To make student demonstrate a sound technical presentation of their selected project topic.
6. To make students aware of different software tools and soft skills required to practice at various stages of project execution.

Course Outcomes:



After completion of the course, students will be able:

1. Apply different design concepts to plan a solution to the problem undertaken
2. Evaluate the performance and detailed analysis of the outcome of the proposed solution for the problem undertaken
3. Work in a project group following work ethics
4. Communicate with engineers and the community at large in written and oral forms
5. Demonstrate the knowledge, skills, and attitudes of a professional engineer.
6. Select and use the proper programming solution, simulator, and necessary soft skills to provide a solution to the problem undertaken.

The objective of Project Phase II is to enable the student to extend further the investigative study taken up under Project I, either fully practical or involving both theoretical and practical work, under the guidance of a supervisor from the department alone or jointly with a Supervisor drawn from the R&D laboratory/Industry. This is expected to provide good training for the student(s) in R&D work and technical leadership.

The assignment normally includes:

1. In-depth study of the topic assigned in light of the report prepared under Project Phase-I
2. Review and finalization of the approach to the problem relating to the assigned topic
3. Preparing an action plan for conducting the investigation, including teamwork;
4. Detailed analysis/modeling/simulation/design/problemsolving/experiment as needed
5. Final development of product/process, testing, results, conclusions, and future directions
6. Preparing a paper for conference presentation/publication in journals, if possible
7. Preparing a project document in the standard format for being evaluated by the department.
8. Final seminar presentation before a departmental committee

Project Report:

The project reports that the following format should be strictly followed.

1. Page size: Trimmed A4
2. Top Margin: 1.00 Inches
3. Bottom Margin: 1.32 Inches
4. Left Margin: 1.5 Inches
5. Right Margin: 1.0 Inches
6. Para Text: Times New Roman 12 point font
7. Line Spacing: 1.5 Lines
8. Page Numbers: Right-aligned at footer, font 12-point Times New Roman
9. Headings: New Times Roman, 14-point, Boldface
10. Index of Report:
 - i) Title Sheet
 - ii) Certificate from Guide/ College
 - iii) Acknowledgement
 - iv) Abstract (Brief content of the work)
 - v) List of Figures
 - vi) List of Table



Report Content:

1. Introduction (History, Importance of Project Area, Problem identification, Objective of the Project)
2. Literature Review
3. Design/ Experimentation / Actual work carried out for the same.
4. Observation/ Analysis/ Findings/Results
5. Discussion on Results and Conclusion

References or Bibliography: References should have the following format

For Books: “Title of Book”; Authors; Publisher; Edition;

